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Topology-Based Recommendation of Users in Micro-Blogging Communities

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Abstract Nowadays, more and more users share real-time news and information in micro-blogging communities such as Twitter, Tumblr or Plurk. In these sites, information is shared via a followers/followees social network structure in which a follower will receive all the micro-blogs from the users he/she follows, named followees. With the increasing number of registered users in this kind of sites, finding relevant and reliable sources of information becomes essential. The reduced number of characters present in micro-posts along with the informal language commonly used in these sites make it difficult to apply standard content-based approaches to the problem of user recommendation. To address this problem, we propose an algorithm for recommending relevant users that explores the topology of the network considering different factors that allow us to identify users that can be considered good information sources. Experimental evaluation conducted with a group of users is reported, demonstrating the potential of the approach.

Keywords social networking, social technology, recommender system

1 Introduction

Micro-blogging activity taking place in sites such as Twitter, Tumblr or Plurk is becoming every day more important as real-time information source and news spreading medium. In the followers/followees social structure defined in these communities, a follower will receive all the micro-blogs from the users he/she follows, known as followees, even though they do not necessarily follow him/her back. In turn, re-tweeting allows users to spread information beyond the followers of the user that post the tweet in the first place.

Since Twitter is the indisputable leader of micro-blogging sites, most of the research in this area is done using this platform. Studies conducted to understand Twitter usage[1-2] revealed that few users maintain reciprocal relationships with other users, which can be regarded as friends or acquaintances, while most of them behave either as information sources or information seekers. Users behaving as information sources tend to collect a large amount of followers as they are actually posting useful information or news. In turn, information seekers follow several users to obtain the information they are looking for and rarely post any tweet themselves.

Finding high quality sources among the expanding micro-blogging community becomes essential for information seekers in order to cope with information overload. In this paper we present a topology-based followee recommendation algorithm aiming at identifying potentially interesting users to follow in micro-blogging networks. This algorithm explores the graph of connections starting at the target user (the user to whom we wish to recommend previously unknown followees), selects a set of candidate users to recommend and ranks them according to a scoring function that favors those users exhibiting the distinctive behavior of information sources.

Unlike other studies that focus on ranking users according to their influence in the entire network[3-4], the algorithm we propose explores the followed/following relationships of the user up to a certain level, so that more personalized factors are considered in the selection of candidates for recommendation, such as the number of friends in common with the target user. Since only the topology of the social structure is used but not the content of tweets, this algorithm also differs from works exploiting user-generated content in micro-

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blogging sites to filter information streams[5-7] or to extract topic-based preferences for recommendation[8].

The rest of this paper is organized as follows. Section 2 introduces some background and reviews related research in the area. Section 3 describes our approach to the problem of followee recommendation in Twitter. Subsections 4.1 and 4.2 present the experiments we performed to validate our proposal and discuss the results obtained. Finally, Section 5 discusses some aspects of our proposal and present our conclusions.

2 Background and Related Work

Answering the question "What are you doing?" using 140 characters or less has become the new phenomenon of the social web. Micro-blogging services have revolutionized the way information is consumed, favoring the brevity of the texts, the mobility of users and virtual networks as an emerging social environment. Micro-blogging services have also emerged as an important source of real-time news updates since the short nature of updates allows users to post news quickly, reaching its audience in seconds.

Among these services, Twitter is the micro-blogging social network with the greatest growth in the last years①. Twitter enables users to send and receive messages with a length shorter than 140 characters that are called "tweets" or status updates. Relationships in Twitter are unidirectional: a Twitter user interested in the tweets published by another user U registers him/herself as a "follower". Although user U has no need to follow his/her followers back, it is possible for him/her to obtain the list of users following him/her.

Several recent research efforts have been dedicated to understand micro-blogging as a novel form of communication and news spreading medium. Java *et al.*[1] and Krishnamurthy *et al.*[2] established four key profiles in the content of entries posted to Twitter:

1) daily trivia,
2) conversations in small communities,
3) information and URLs sharing,
4) news and views dissemination.

With regard to user profiles, the same studies point to three main categories:

1) *Information Sources*: users that are characterized by having a much larger number of followers than they themselves are following and by posting frequent updates. This category includes users that *retweet*② relevant posts with information about different subjects, such as sports, movies, music or about a particular rock band.

2) *Friends or Acquaintances*: family, colleagues and contacts from other social networks in which the user participated previously.

3) *Information Seekers*: mainly readers of the contributions of others, but who rarely post tweets authored by themselves. These users use Twitter to get information on particular subject, as a form of RSS (Really Simple Syndication) reader, registering themselves as followers of their favorite artists, celebrities, bloggers, or TV programs.

For this last type of users finding high quality and reliable information sources in the constantly increasing Twitter community becomes a challenging issue.

The problem of helping users to find and to connect with people on-line to take advantage of their friend relationships has been also studied in the context of general social networks. For example, SONAR[9] recommends related people in the context of enterprises by aggregating information about relationships as reflected in different sources within an organization, such as organizational chart relationships, co-authorship of papers, patents, projects and others. Liben-Nowell and Kleinberg[10] presented different methods for link prediction based on node neighborhoods and on the ensemble of all paths. These methods were evaluated using co-authorship networks. Authors found that there is indeed useful information contained in the network topology alone. Chen *et al.*[11] compared relationship-based and content-based algorithms in making people recommendations, finding that the first ones are better at finding known contacts whereas the second ones are stronger at discovering new friends. Weighted minimum-message ratio (WMR)[12] is a graph-based algorithm which generates a personalized list of friends in a social network built according to the observed interaction among members. Unlike these algorithms that gather social networks in enclosed domains, mainly starting from structured data (such as interactions, co-authorship relations), we propose a user recommendation algorithm that takes advantage of Twitter social structure populated by massive, unstructured and user-generated content.

The influence of users in Twitter has also been the subject of several studies. In [13] it was shown that ranking users by the number of followers and by their PageRank give similar results. However, ranking users by the number of re-tweets indicates a gap between influence inferred from the number of followers and that from the popularity of users' tweets. Coincidentally, a

①In 2010 Twitter grew by more than 100 million registered accounts (http://yearinreview.twitter.com/whosnew/). Accessed on August 2011.

②In Twitter jargon, a *retweet* is how Twitter users share interesting tweets from the people they are following with their own followers. To give credit to the original person, users usually put *"RT"* plus the originator's username at the beginning of the tweet.

comparison between in-degree, re-tweets and mentions as influence indicators[14] concluded that the first is more related to user popularity. Analyzing spawning retweets and mentions to other users in Twitter, it was found that most influential users hold significant influence over a variety of topics but this influence is gained only through a concentrated effort (such as limiting tweets to a single topic). TwitterRank[3], an extension of PageRank algorithm, tries to find influential twitterers by taking into account the topical similarity between users as well as the link structure. TURank[4] considers the social graph and the actual tweet flow. Garcia and Amatriain[15] proposed a method to weight popularity and activity of links for ranking users. User recommendation, however, cannot be based exclusively on general influence rankings studies on the complete Twittersphere since people who are popular in Twitter would not necessarily match a particular user's interests. For example, if a user follows accounts talking about technology, he/she would not be interested in Ashton Kutcher, one of the most influential Twitter accounts according to [13].

While the mentioned studies focus on analyzing micro-blogging usage, other works try to capitalize the massive amount of user-generated content as a novel source of preference and profiling information for recommendation. Chen *et al.*[5] proposed an approach to recommend interesting URLs coming from information streams such as tweets based on two topic interest models of the target user and a social voting mechanism so that the most popular URLs within the group are recommended. Buzzer[6] indexes tweets and recent news appearing in user specified feeds, which are considered as examples of user preferences, to be matched against tweets from the public timeline or from the user Twitter friends for story ranking and recommendation. Esparza *et al.*[7] addressed the problem of using real-time opinions of movie fans expressed through short textual reviews for recommendation. This work assumes that micro-posts carry on preference-like information that can be used in content-based and collaborative filtering recommendation. More recently, Abel *et al.*[16] introduced a framework for user modeling on Twitter which enriches the semantics of tweets by identifying topics and entities in the context of news recommendations.

In contrast to the previous studies that address the problem of suggesting potentially relevant content from micro-blogging services, we concentrate on recommending interesting people to follow. In this direction, Sun *et al.*[17] proposed a diffusion-based micro-blogging recommendation framework which identifies a small number of users playing the role of news reporters and recommends them to information seekers during emergency events. Closest to our work are the algorithms for recommending followees in Twitter evaluated and compared using a subset of users in [8]. Multiple profiling strategies were considered according to how users are represented in a content-based approach, a collaborative filtering approach and two hybrid approaches. User profiles are indexed and recommendations generated using a search engine, receiving a ranked-list of relevant Twitter users based on a target user profile or a specific set of query terms. Our work differs from this approach in that we do not require indexing profiles from Twitter users, instead topology-based and content-based algorithms explore the follower/followee network in order to find candidate users to recommend. Furthermore, we consider in the evaluation of our approach the target user assessment about the his/her interest in the provided recommendations.

The main difference between existent work and our work is that the mentioned approaches for followee recommendations, except for the approach presented in [8], were evaluated using datasets gathered from Twitter, with no assessment about the target user's interest in the recommendations. In other words, the target user's interest in a followee recommended that is not in the current list of the target user's followees cannot be assessed within these datasets in order to determinate the correctness of the recommendation. For this reason, the approach proposed in this work is evaluated with a controlled experiment with real users.

3 Followees Recommendations in Follower/Followee Networks

In contrast to most work on Twitter that focus on exploiting the content of tweets[5-6,16,18], we take advantage of the well studied characteristics of Twitter network. Particulary, we consider the characterization of users acting as information sources and information seekers. The algorithm we propose for recommending users in follower/followee networks consists of two steps: 1) we explore the target user's neighborhood in search of candidates and 2) we rank candidates according to different weighting features. These steps are detailed in Subsections 3.1 and 3.2, respectively.

3.1 Finding Candidates

The general idea of the algorithm we implemented is to suggest users that are in the neighborhood of the target user, where the neighborhood of a user is determined from the follower/followee relations in the social network.

In order to find candidate followees to recommend to a target user U, we base our search algorithm on the following hypothesis: the users followed by the followers of U's followees are possible candidates to recommend to U. In other words, if a user F follows a user that is also followed by U, then other people followed by F can be interesting to U.

The rationale behind this hypothesis is that the target user is an information seeker that has already identified some interesting users acting as information sources, which are his/her current followees. Other people that also follow some of the users in this group (i.e., is subscribe to some of the same information sources) have interests in common with the target user and might have discovered other relevant information sources in the same topics, which are in turn their followees.

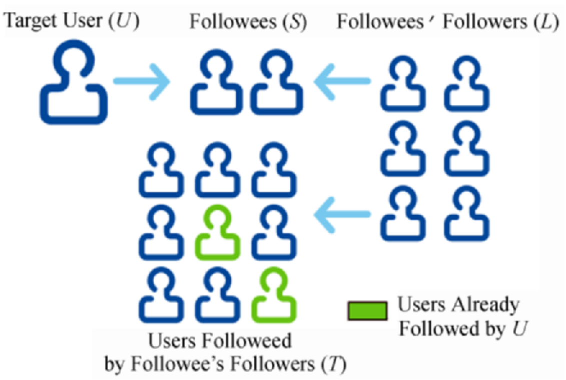

Fig.1. Scheme for finding candidate recommendations.

This scheme is outlined in Fig.1 and can be resumed in the following steps:

1) Starting with the target user U, we first obtain the list of users he/she follows, let call this list S.

$$S = \bigcup_{s \in followees(U)} s.$$

2) From each element in S we get its followers, let call the union of all these lists L.

$$L = \bigcup_{s \in S} followers(s).$$

3) From each element in L, we get its followees to obtain the list of possible candidates to recommend. Let call the union of all these lists T.

$$T = \bigcup_{l \in L} followees(l).$$

4) Exclude from T those users that the target user already follows. Let call the resulting list R.

$$R = T - S.$$

Each element in R is a possible user to recommend to the target user. Notice that each element can appear more than once in R, depending on the number of times that each user appears in the followees or followers lists obtained at steps 2 and 3 above.

3.2 Weighting Features

Once we find the list R of candidate recommendations for the target user, we explore different features to give a score to each unique user $x \in R$. The first feature explored is the relation between the number of followers a user has with respect to the number of users the given user follows, as shown in (1).

$$w_f(x) = \frac{|i \in followers(x)|}{|i \in followees(x)|}. \quad (1)$$

Since we seek for sources of information to recommend, we assume that this kind of users will have a lot of followers and that they will follow few people. If user x has no followees, then only the number of followers is considered without changing the significance of the weighting feature.

We use this metric as a baseline for comparison with other metrics. Our aim is to demonstrate that metrics for ranking popular users on Twitter are not good for ranking recommendations of users that a target user might be interested in following. In [13] it has been shown that the rankings of users that can be obtained by number of followers and by PageRank[19] are very similar. We opt to use this factor as an estimator of the "importance" of a given user because the number of followers is a metric by far more easily to obtain that the user PageRank in a network with an order of almost 2 billion social relations.

The second feature explored corresponds to the number of occurrences of the candidate user in the final list of $|R|$ candidates for recommendations, as shown in (2).

$$w_o(x) = \frac{|\{i \in R\, s.t.\, i = x\}|}{|R|}. \quad (2)$$

The number of occurrences of a given user x in this final list is an indicator of the amount of (indirect) neighbors that also have x as a (direct) connection itself. The third feature we consider is the number of friends in common between the target user U and the candidate recommendation x:

$$w_c(x) = |i \in followees(x) \bigcap i \in followees(U)|. \quad (3)$$

Finally, we consider two combinations of these features: the average of the three features, and their product:

$$w_s(x) = \frac{1}{3}[w_o(x) + w_f(x) + w_c(x)], \quad (4)$$

$$w_p(x) = w_o(x) \times w_f(x) \times w_c(x). \quad (5)$$

It is worth noticing that the selection of these weighting features is not arbitrary. Our choice is based on a deep analysis of previous studies about Twitter and particular properties of this specific network that makes general link prediction approaches unsuitable. All the studies about the properties of the Twitter network agree in that there is a minimal overlap with the features available on other OSNs (Online Social Networks).

4 Experimental Evaluation

To evaluate the proposed algorithm, we performed two experiments. First, in Subsection 4.1 we describe an online experiment with real users in which we evaluate the real interest of the target user in the followees recommended by our algorithm. Then, in Subsection 4.2 we describe an offline experiment performed using a subgraph of the Twitter network crawled using the participant users' accounts as seeds. In this second experiment we hid a percentage of the users already being followed by the target users and evaluated if our algorithm was able to rediscover those hidden connections using the remaining connections.

4.1 Live-User Experiment

We carried out a preliminary experiment using a group U_{test} consisting of 14 users. These users, 8 males and 6 females, were in the last years of their course of studies and were students of a Recommender Systems related course dictated at our University as an elective course during 2010. The students selected for the experiment were volunteers familiarized with Twitter.

During the first part of the course, we asked these users to create a Twitter account and to follow at least 20 Twitter users who published information or news about a set of particular subjects of their interest. The general interests expressed by users ranged between diverse subjects such as technology, software, math, science, football, tennis, basketball, religion, movies, journalists, government, music, cooking, shoes, TV programs and even other students in their faculty. Some users only concentrated on one particular subject while others distributed their followees among several topics.

During the second part of the course, we provided these users with a desktop tool that allowed them to login to Twitter and ask for followees recommendations. Since the users who participated in the experiment were students of a Recommender Systems course, all of them had knowledge about concepts such as rankings and metrics. As part of a not compulsory practical exercise of the course they were motivated to discover which metric better ranked recommendation results and to write a brief report about the results they obtained for their particular case. The desktop application provided for this exercise allowed students to select the weighting feature by which they liked to rank recommendations, with no predefined order.

In all cases, 20 recommendations were presented to the users. Then, we asked the students to explicitly evaluate whether the recommendations were relevant or not according to the same topical criteria they had chosen to select their followees as information sources in the first place. For each recommendation in the resulting ranking the application showed the user name, description, profile picture and a link to the home page of the corresponding account. This link could be used to read the tweets published by the recommended user in the case that the information provided by the application was not enough to determine the student's interest in the recommendation. The question we asked students to ask themselves to determine whether a recommendation was relevant or not was "Would you have followed this recommended user in the first place (when selecting which users to follow in the first part of the experiment), if you had known this account?" For example, if a given student was interested in technology and he/she had not discovered the account @TechCrunch during his/her first selection of followees, that would be an interest recommendation because @TechCrunch tweets about news on technology.

The quality of lists of top-N followee recommendations generated for each user who participated in the experiment was evaluated considering the standard precision at different positions of the ranking:

$$\mathrm{P@}k = \frac{|\{\text{relevant users with rank } k \text{ or less}\}|}{k}. \quad (6)$$

Thus, precision at rank k (P@k) is defined as the proportion of recommended followees that are relevant at a given cut-off rank. For example, P@5 ("precision at five") is defined as the percentage of relevant recommendations among the first five, averaged over all runs. In the reported experiments we evaluated precision for values of k equal to 1, 5, 10 and 20, although k values of 1 and 5 are the most common sizes for recommendation lists reported in the literature as people tend to pay more attention to the first few results that are presented. Fig.2 shows the precision achieved by the algorithm, averaged between all users, for each weighting feature.

We can observe several interesting facts in the results presented in Fig.2. First, it shows that $w_o(x)$, the weighting feature considering the number of occurrences of a user in the list of recommendations as gathered by the algorithm proposed, generates better precision scores than any other weighting feature explored. For this weighting feature we obtained a good recommendation in the first position of the ranking for 93% of the users. For longer ranking lists, precision decreases from 0.73 for P@5 to 0.64 for P@20, which we believe are all good results.

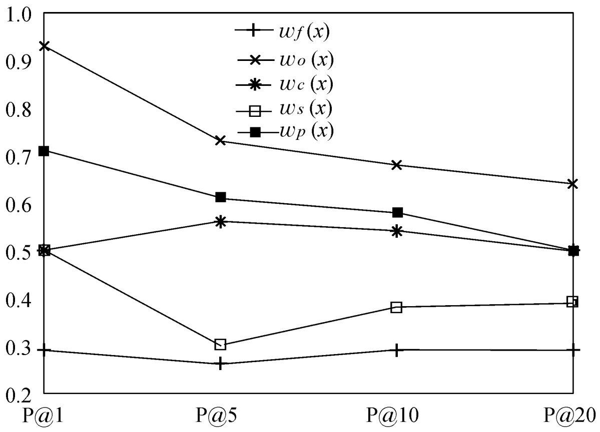

Fig.2. Average precision for each weighting feature.

It is worth noticing that although we reported results up to P@20, recommendations lists tend to be shorter (frequently 5) in order to help the user to focus on the most relevant results. In these small lists the algorithm reached good levels of precision, recommending mostly relevant users.

The weighting feature considering the number of followers, $w_f(x)$, got the worst precision scores, with values under 30%. This fact reveals that this metric, although widely used in other approaches as mentioned in Section 2, is only good at measuring a user's general popularity in the entire Twitter network, but popularity does not necessarily translate into relevance for a particular user. Celebrities and politicians, such as Barack Obama (@barackobama), Lady Gaga (@ladygaga), Yoko Ono (@yokoono), and Tom Cruise (@tomcruise) were a common factor in the rankings of many users regardless their particular interests. Among other popular users suggested that in some cases met the user's interests were popular blogs and news media such as Mundo Geek (@mundo_geek), C5N (@C5N), El Pais (@el_pais), Mundo Deportivo (@mundodeportivo), Red Hat News (@redhatnews) and Fox Sport LA (@foxsportslat).

A similar situation occurs with $w_c(x)$, the weighting feature considering the number of friends in common between the target user U and the candidate user to recommend to U. Although precision is better than $w_f(x)$ for every size of the recommendation lists, this weighting feature does not reach the performances obtained with $w_o(x)$. This result is logic since the fact that two users U and X share a friend Y does not necessarily means that X is a good information source.

We also found that $w_s(x)$ tends to perform poorly. This score is affected by the term corresponding to the relation between the number of followers and the number of followees, which in most cases is higher than the other terms involved. This factor highly affects the overall average among the three weighting features, causing a decrease in precision.

The second score which combines the three weighting features, $w_p(x)$, seems to overcome this problem since in this case each weighting feature is multiplied to obtain the final score. Nevertheless, celebrities and very popular Twitter user accounts also tend to appear at the top positions of the ranking diminishing the general precision again. However, the factor corresponding to $w_o(x)$ also makes good recommendations to appear interleaved with some popular users on Twitter.

Another interesting issue observed in the results presented in Fig.2 is that for both $w_f(x)$ and $w_c(x)$ precision tends to keep almost constant across different sizes in the list of recommendations and even with a slightly increment as the size of the recommendation set increases. This fact seems to contradict the definition of precision in the information retrieval sense which, by principle, should decrease as the number of recommendations increases. However, this behavior occurs because all $w_f(x)$, $w_s(x)$ and $w_c(x)$ do not concentrate relevant recommendations on the top positions of the ranking. On contrary, we can observe that for $w_o(x)$ and $w_p(x)$ relevant recommendations tend to be clustered towards the top of the ranking.

Although precision measure gives a general idea of the overall performance of the presented weighting features, it is also very important to consider the position of relevant recommendations in the ranking presented to the user. Since it is known that users focus their attention on items at the top of a list of recommendations[20], if relevant recommendations appear at the top of the ranking using one algorithm and at the bottom of the ranking using the other, the first algorithm will be perceived as better performing by users even though their general precision might be similar.

Success at rank k (S@k) is a metric commonly used for ranked lists of recommendations. The success at rank k is defined as the probability of finding a good recommendation among the top k recommended users. In other words, S@k is the percentage of runs in which there was at least one relevant user among the first k recommended users. Fig.3 shows the results we obtained for this metric with values of k ranging from 1 to 10.

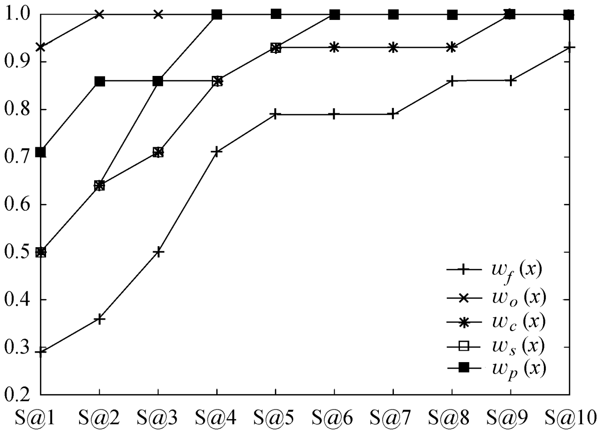

Fig.3. Success at rank k for each weighting feature.

S@1 is equivalent to P@1 by definition. Then, we can see that scoring users with $w_o(x)$ always positions relevant users above in the ranking than the other weighting features. The ranking according to $w_p(x)$ allows users to find a relevant recommendation always at the most at position 4 in the ranking, seconded by $w_s(x)$ for which we obtain full success at position 6. With this metric we can confirm that $w_f(x)$ and $w_c(x)$ are not good weighting factors by their own.

Discounted cumulative gain (DCG) is another measure of effectiveness used to evaluate ranked lists of recommendations. DCG measures the usefulness, or gain, of a document based on its position in the result list using a graded relevance scale of documents in a list of recommendations,. The gain is accumulated from the top of the result list to the bottom with the gain of each result discounted at lower ranks. The premise of DCG is that highly relevant documents appearing lower in a list should be penalized as the graded relevance value is reduced logarithmically proportional to the position of the result. The DCG accumulated at a particular rank position k is defined as $\text{DCG@}k = rel_1 + \sum_{i=2}^{k} \frac{rel_i}{\log_2 i}$, where rel_i indicates whether the recommended user at position i in the ranking is relevant or not (taking value 1 or 0, respectively). DCG is often normalized using an *ideal DCG* that is computed by sorting documents of a result list by relevance. Fig.4 shows the normalized DCG obtained for both algorithms at four different positions of the ranking: nDCG@1, nDCG@5, nDCG@10 and nDCG@20.

To study further the algorithm ability to rank followees for recommendation, we measure the effectiveness of the ranking in terms of mean average precision (MAP), which is computed as shown in (7):

$$\text{MAP} = \frac{\sum_{u=1}^{|U_{test}|} \text{AveP}(u)}{|U_{test}|}, \tag{7}$$

where $|U_{test}|$ is the number of users involved in the experiment.

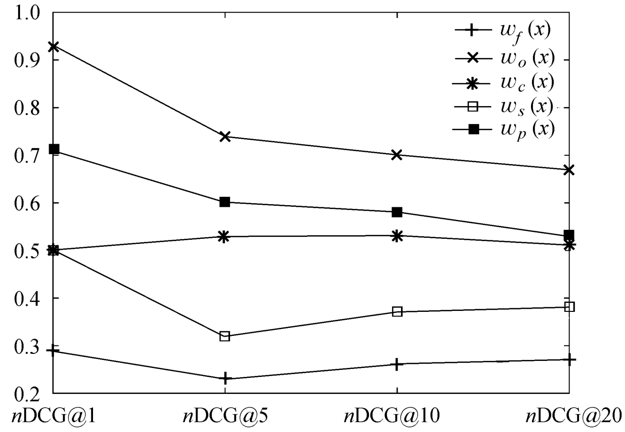

Fig.4. Normalized discounted cumulative gain.

Precision is a single-value metric based on the whole list of recommendations presented to the user. Average precision (AveP) emphasizes ranking relevant users higher. It is the average of precisions computed at the point of each of the relevant users in the ranked sequence:

$$\text{AveP} = \frac{\sum_{k=1}^{N}(\text{P@}k \times rel(k))}{\text{number of relevant users}},$$

where N is the number of recommendations, $rel(k)$ a binary function on the relevance of the user recommended at position k in the ranking, and P@k is the precision at position k as defined in (6).

Fig.5 plots the mean average precision obtained using the different weighting features.

This metric gives us another view of which weighting feature generates better ranking of recommendations. We confirm that $w_o(x)$ always ranks users better than the other proposed weighting features, while ranking users by their "popularity" does not generate good recommendations.

The experiments presented make us believe that there is reason to be optimistic about the potential for a followee recommender for Twitter using the method described in Subsection 3.1 to obtain a list of candidates and simply ranking them by the number of occurrences of each candidate in the list generated by this method. Among the advantages of this method when compared

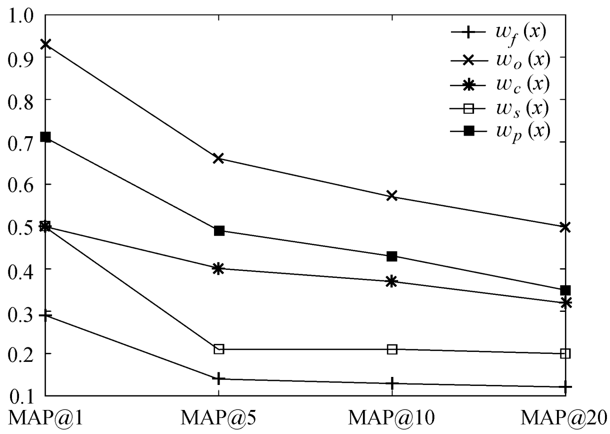

Fig.5. Mean average precision.

with content-based alternatives is that recommendations can be found quickly based on a simple analysis of the network structure, without considering the content of the tweets posted by the candidate user. Nevertheless, we also believe that combining the proposed method with an analysis of the content of the tweets posted by a user in the list of candidates can improve the precision of a followee recommender system, at the expense of computational performance.

From the related work presented in Section 2, the approach that we find more similar to ours (and the only one, up to our knowledge that experimented with real users in a controlled experiment) is Twittomender, proposed by Hannon *et al.*[8]. Although the results presented in this work by Hannon *et al.* are not fully comparable to the results presented in this article since different datasets were used, next we present a comparison about the precision reported for Twittomender and the precision obtained with our approach.

Twittomender creates different indexes for all users in the dataset generated from different sources of profile information. Four of these indexes are content-based, modeling users by their own tweets, by the tweets of their followers, by the tweets of their followees and by a combination of the three. The three remaining strategies are topology-based and modeling users by the IDs of their followees, by the IDs of their followees and by a combination of both.

The strategy used for ranking users in the online experiment presented in [8] generates the seven rankings according to the different approaches described above and then generates a single ranking by merging those seven rankings. When merging the rankings a scoring function is used based on the position of each user in the recommendation lists. In this way users that are frequently present in high positions are preferred over users that are recommended less frequent or in lower positions.

Hannon *et al.* performed a live user trial with 34 users, reporting a precision of about 38.2% for $k = 5$ and 33.8% for $k = 10$. Table 1 summarizes the comparison between Twittomender and our system. Notice that values for Twittomender system are approximate because they were taken (and in some cases computed) from the graphics presented in the article.

Table 1. Comparisions between Twittomender and Our Approach

	Twittomender "Live-User" Trial	Our Approach
No. Seeds	34	14
No. Users	100 000	1 443 111
P@5	~38.2%	72.9%
P@10	~33.8%	67.9%
P@20	~26.9%	64.3%

It is worth noticing that although the number of volunteers who participated in Twittomender experiment is more than twice the number of volunteers who participated in our experiment, the number of Twitter users involved in our experiment is by far larger than the number of users in their database. Furthermore, Twittomender can only recommend users that are previously indexed. When a user is registered into the system, all his/her followees and followers profiles along with his/her own profile are indexed. Our work differs from this approach in that we do not require indexing profiles from Twitter users; instead a topology-based algorithm explores three levels of the follower/followee network in order to find candidate users to recommend.

4.2 Off-Line Experiment

To complement the experimentation with real users, we carried out an evaluation of the approach in an off-line scenario. For this experiment we crawled a sub-graph of the Twitter network corresponding to three levels of both followee and follower relations, centered on each user in U_{test}, that is using the user IDs of the user accounts created by the students as seeds. The resulting dataset consists of 1 443 111 Twitter users and 3 462 179 following relations already existing among them. Figs.6(a) and 6(b) show the distribution of the number of followers and followees, respectively, for users in the crawled sub-network. It can be observed in these figures that both distributions follow a power law.

Once the social graph was obtained, the off-line experiments were carried out using a holdout strategy in which some connections of the target users' followees were hidden. Then we applied our recommendation algorithm to then verify if it was able to discover and suggest the hidden connections as potential followees. In all the experiments, the set of followees of each user was partitioned into 70% for training and 30% for testing.

The followees selected for testing, denoted as U_{test}, correspond to the hidden connections that we then tried to rediscover starting with the remaining 70% connections to search and evaluate candidate recommendations using our algorithm. If followees in the 30% group were suggested to the target user in spite of had being concealed, it means that the algorithm is able to locate these users through the 70% non-concealed followees and their relationships. In order to make the results less sensitive to the particular training/testing partitioning of the followees, in all experiments the average and standard deviation of 5 runs for each individual user are reported, each time using a different random partitioning into training and test sets.

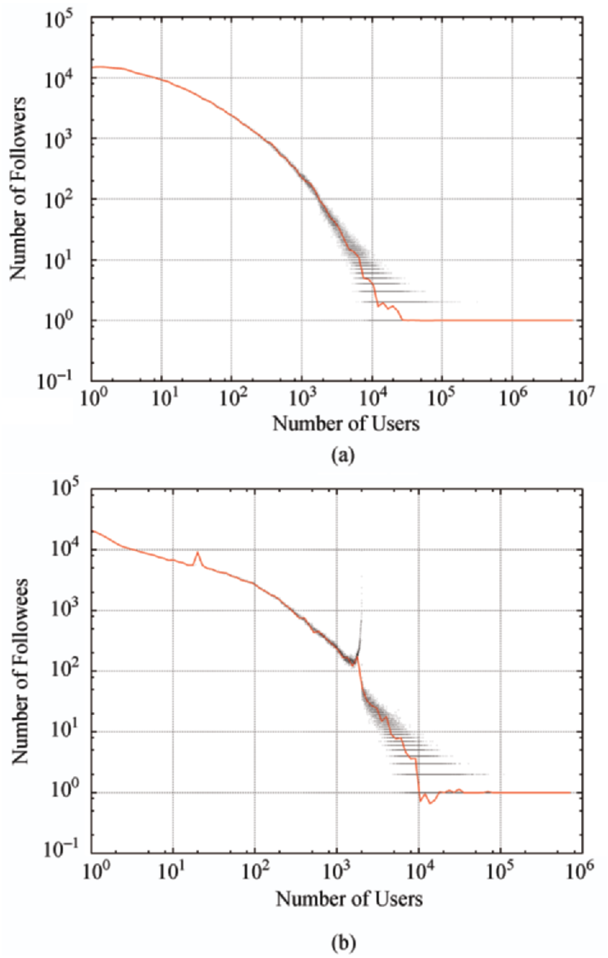

Fig.6. Distributions of the number of followers and followees for users in the dataset.

The top-N followee recommendations generated for the group of users used were evaluated using the previously mentioned metrics of precision, mean average precision (MAP) and success. In this case, precision measures the average percentage of overlap between a given recommendation list and the user actual list of followees. Only the w_o feature was considered as it was the best performing in the live-users experiment.

The number of hits in a recommendation list, which is the number of followees in the test set that were also present in the top-N recommended followees for a given test user was also reported in these experiments. If $|U_{test}|$ is the total number of testing users, the hit-rate (HR) of the recommendation algorithm is computed as[21]:

$$\mathrm{HR} = \frac{\text{number of hits}}{|U_{test}|}. \tag{8}$$

HR grants high values to an algorithm if it is able to predict the followees in the test sets of the corresponding users, while assigns low values to the algorithm if it is not able to recommend the hidden followees.

One limitation of this measure is that it treats all hits equally regardless of where they appear in the list of the top-N recommended items. Average reciprocal hit-rank (ARHR) rewards each hit based on where it occurs in the top-N followees that were recommended by a particular strategy (RE). If h is the number of hits that occur at positions $p_1, p_2, \ldots, p_h$ within the top-N lists (i.e., $1 \leqslant p_i \leqslant N$), then the average reciprocal hit-rank is equal to:

$$\mathrm{ARHR(RE)} = \frac{1}{|U_{test}|} \sum_{i=1}^{h} \frac{1}{p_i}. \tag{9}$$

That is, hits that occur earlier in the top-N lists are weighted higher than hits that occur later in the list. The highest value of ARHR is equal to the hit-rate and occurs when all the hits occur in the first position, whereas the lowest value of the ARHR is equal to hit-rate$/N$ when all the hits occur in the last position in the list of the top-N recommendations.

Fig.7 shows the results in terms of hit-rate and average reciprocal hit rank. It can be observed that the

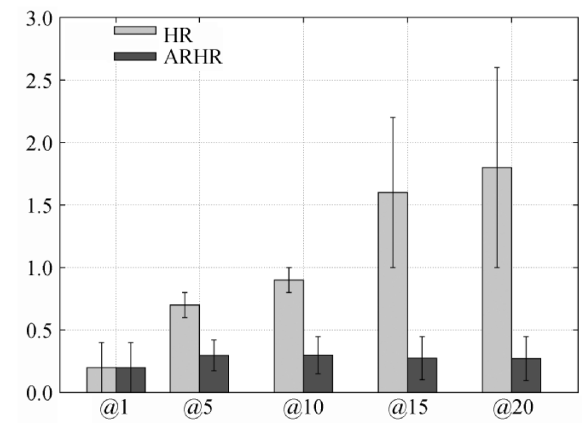

Fig.7. Hit rate and ARHR at rank k for each weighting feature.

algorithm was able of discovering some of each user's followees again despite being hidden and recommending them to users. Fig.8 shows precision and mean average precision for the same experiments. While precision diminishes for longer recommendation lists MAP value maintains constant, indicating that hits are at top positions. Fig.9 shows success of recommendation at different ranking positions showing that at least a followee is rediscovered in most cases.

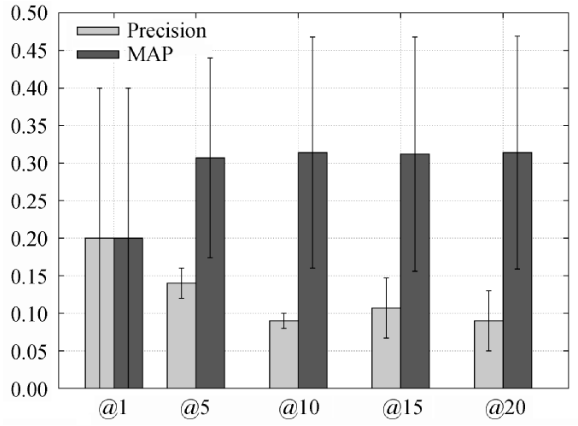

Fig.8. Precision and MAP at rank k for each weighting feature.

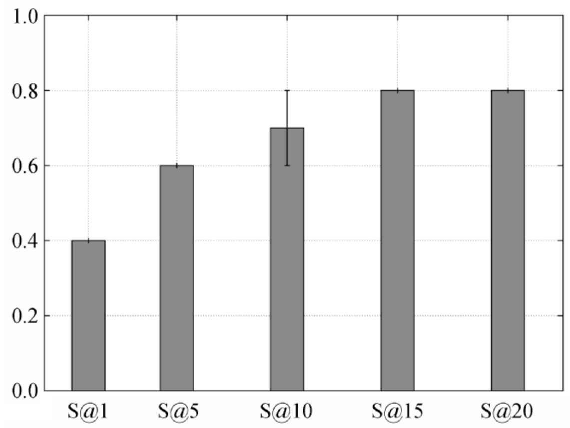

Fig.9. Success at rank k for each weighting feature.

It is worth noticing that in the previous results the effectiveness of the algorithm to identify followees is been underestimated given the testing methodology employed. Users suggested to the target user that are not in the test set are not necessarily uninteresting, although they are considered incorrect recommendations in the calculation of the precision and hit-rate metrics. This aspect was considered in the live user experiment in which the users inspected each recommended followee and judged their relevance.

5 Discussion and Conclusions

In this article we presented a simple but effective algorithm for recommending followees in the Twitter social network. This algorithm first explores the target user neighborhood in search of candidate recommendations and then sorts these candidates according to different weighting features: the relation between the number of followers and the number of followees, the number of occurrences of each candidate in the final list, the number of friends in common, and two combinations of the three features.

We evaluated the proposed algorithm with real users and we obtained satisfactory results in finding good followee recommendations. We found that considering just the overlapping users among the different lists of follower and followees explored by our crawling method gives better results than the other features considered. As expected, the indegree of a user is not a good feature for ranking followee recommendations. Considering the number of followers for ranking users put celebrities and popular Twitter accounts at the top of the list, but these recommendations are not necessarily interesting for a particular user. However, there are some interesting recommendations discovered by this feature, such as top bloggers who write about a particular subject or news media accounts.

Although the results reported seem promising, we are planning to repeat the experiment this year in order to involve more users in the experiment and obtain more statistical support for the results reported. Moreover, we are very optimistic about the potential improvements that we can obtain by extending the presented approach with content-based techniques. A natural extension of our approach in which we are currently working on is a hybrid algorithm that filters the candidate recommendations found with the topology-based method with a content-based analysis of the tweets posted by the users. In this new approach, a target user U is modeled with a vector of terms built from a content analysis of the tweets posted by U's followees. This vector is then compared with the vector of terms corresponding to each candidate recommendation and the similarity obtained is considered in the generation of the ranking.

The results reported in this article make us feel really enthusiastic about the potentials of Twitter for building recommender systems of sources of information.

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